

1. FDA Expedited Programs Framework

1.1. Definitions

1.1.1. Fast Track (FT)

Definition:

FT is an FDA designation designed to accelerate the development and review of drugs intended to treat serious conditions and address unmet medical needs.

Eligibility Criteria:

The drug must meet two key requirements: it must target a serious condition and there must be an unmet medical need.

A serious condition refers to a disease or condition that causes functional impairment and has a substantial impact on day-to-day functioning – generally, these are serious or life-threatening illnesses.

An unmet medical need exists when the disease or condition is not adequately addressed by available therapy. This includes situations where existing treatments cannot address a serious outcome of the condition, or where a drug could address an emerging or anticipated public health need. In addition, if the only available therapy was approved under accelerated approval based on a surrogate or intermediate clinical endpoint, and its clinical benefit has not yet been verified, that situation also qualifies as an unmet medical need.

Regulatory Advantages:

First, sponsors can have frequent meetings and communications with the FDA, obtaining early regulatory feedback, speeding up information exchange, and facilitating problem-solving.

Second, Rolling Review. Sponsors can submit NDA/BLA sections piecemeal, FDA reviews as it receives, thereby shortening the overall review timeline. Moreover,

products that receive Fast Track designation often have an easier path to qualifying for even more expedited pathways.

Typical Endpoints:

Depending on the disease and stage of development, Fast Track allows the use of surrogate endpoints (e.g., objective response rate, ORR), intermediate clinical endpoints (e.g., duration of response, DoR), and ultimate clinical endpoints (e.g., overall survival, OS). The key principle is that the chosen endpoint must be reasonably likely to predict clinical benefit.

1.1.2. Breakthrough Therapy Designation (BTD)

Definition:

BTD is an FDA designation intended to expedite the development and review of a drug for treating a serious or life-threatening condition where there is an unmet medical need.

Eligibility Criteria:

The drug must target a serious or life-threatening condition; it must address an unmet medical need; and there must be preliminary clinical data showing that the therapy offers a substantial improvement over available treatments.

Regulatory Advantages:

Products granted BTD benefit from deeper FDA engagement and guidance, including frequent meetings, communications, and advice throughout the development process. Rolling review is also available. In addition, the FDA will assign a project lead to coordinate the internal review team, ensuring smooth communication and efficiency during drug development.

Typical Endpoints:

BTD often relies on surrogate or intermediate clinical endpoints – such as ORR, DoR,

or progression-free survival (PFS). Data supporting BTD can come from single-arm trials; a randomized controlled design isn't always necessary. The key is that the data must convincingly demonstrate a substantial improvement over existing therapies.

1.1.3. Accelerated Approval (AA)

Definition:

Accelerated Approval is an FDA pathway designed for drugs that treat serious conditions and address unmet medical needs.

Eligibility Criteria:

The drug must be intended for a serious disease, and its effectiveness must be reasonably assessed through a surrogate or intermediate endpoint. That prediction needs to have solid scientific backing.

Regulatory Advantages:

AA helps get the drug to patients faster, cutting down wait times. At the same time, the FDA continues oversight after approval. Post-marketing confirmatory trials are required to verify the drug's actual clinical benefit, and if that benefit isn't confirmed, the approval can be withdrawn.

Typical Endpoints:

Commonly used endpoints include tumor response rate (ORR), biomarkers, enzyme activity, and other surrogate measures that can reasonably predict efficacy. These endpoints allow a drug's potential clinical value to be assessed in a shorter timeframe.

1.1.4. Priority Review (PR)

Definition:

PR is a designation established by the FDA to expedite the review of a drug application. It does not change the drug's development pathway or the evidence

requirements.

Eligibility Criteria:

The drug must be intended for a serious condition. The drug must demonstrate a significant improvement in safety, effectiveness, diagnosis, or prevention.

Regulatory Advantages:

The main advantage of PR is a significantly shortened FDA review timeline, typically 6 months. The FDA assigns a dedicated review team to the application, improving communication and review efficiency. PR is often combined with BT or AA, giving sponsors the combined benefits of accelerated development and a faster path to market.

Typical Endpoints:

Priority Review does not prescribe any specific endpoint type, the endpoints depend entirely on the drug itself and the specific NDA/BLA application.

1.1.5. Orphan Drug Designation (ODD)

Definition:

Orphan Drug Designation is an FDA designation created to encourage the development of drugs for rare diseases.

Eligibility Criteria:

The drug must be intended for a rare disease – meaning the disease affects fewer than 200,000 people in the United States. ODD does not require an unmet medical need, nor does it impose hard requirements on disease severity or drug cost. It applies to drugs, biologics, medical devices, and medical foods.

Regulatory Advantages:

Drugs with ODD receive 7 years of market exclusivity after approval – during that

time, the FDA will not approve another identical drug for the same indication, offering strong commercial protection. Sponsors can claim a tax credit equal to 50% of clinical trial costs and are exempt from NDA/BLA application fees. Additionally, government grant funding is available in Phase I/II/III trials. More importantly, ODD can be stacked with other expedited programs.

Typical Endpoints:

ODD does not mandate any specific endpoints. Clinical trial design is more flexible – single-arm trials, lack of placebo control, and surrogate endpoints are acceptable, as long as they reasonably support the drug's safety and effectiveness.

2. Application to Dato- DXd

2.1. Unmet Medical Need Analysis

Dato- DXd partially addresses the unmet medical need in this patient population.

According to the FDA approval summary (U.S. Food and Drug Administration, 2024a). HR+/HER2– metastatic breast cancer is a serious and life-threatening disease. All currently available treatments are palliative, so there is a clear unmet medical need. The standard of care is a CDK4/6 inhibitor combined with endocrine therapy; after progression, patients receive sequential endocrine-based regimens followed by single-agent chemotherapy. However, existing chemotherapy offers limited efficacy, median progression-free survival is only about 4–5 months and is associated with significant hematologic toxicity. In the chemotherapy arm, the incidence of grade 3–4 neutropenia was as high as 35%. Patients tolerate chemotherapy poorly, so new treatment options are needed.

As a Trop-2-directed antibody-drug conjugate, Dato- DXd significantly prolongs PFS and improves ORR and DoR compared with chemotherapy. Although OS did not show a benefit, the FDA considered that the results did not demonstrate a clear trend toward harm. More importantly, Dato- DXd avoids the severe hematologic

toxicities commonly seen with chemotherapy, offering an alternative for patients who cannot tolerate or wish to avoid such toxicity.

While Dato-DXd does introduce new safety risks, and patients reported slightly worse overall side-effect bother compared to chemotherapy (page 5), the FDA judged the overall benefit-risk profile to be acceptable and ultimately approved the drug. Therefore, even though Dato-DXd does not extend overall survival, it does partially address the unmet medical need in this population.

The details of data in table1

Endpoint / Adverse Event	Dato- DXd	Chemotherapy (ICC)
Median PFS (months)	6.9	4.9
PFS hazard ratio (HR)	0.63 (95% CI 0.52–0.76)	—
Objective response rate (ORR)	36%	23%
Median DoR (months)	6.7	5.7
Median OS (months)	18.6	18.3
OS hazard ratio (HR)	1.01 (95% CI 0.83–1.22)	—
Grade 3- 4 neutropenia	1.6%	35%
Any- grade stomatitis	59%	17%
Any- grade dry eye	27%	13%
Any- grade keratitis	24%	10%
Interstitial lung disease (ILD)	4.2%	0%

Table 1 Comparison of Key Outcomes from the TROPION-Breast01 Trial.

2.2.FT and BT Eligibility Assessment

2.2.1. FT Eligibility

Dato- DXd meets the criteria for FT.

HR+/HER2– metastatic breast cancer qualifies as a serious condition with an unmet medical need. In early development, Dato- DXd had already demonstrated Trop- 2- directed antitumor activity and an acceptable safety profile. The subsequent TROPION- Breast01 trial further confirmed the drug's advantages over chemotherapy in terms of PFS and ORR (Table 1). Therefore, Dato- DXd fully satisfies the Fast Track criteria, and a sponsor applying during Phase I/II would likely receive this designation.

2.2.2. BTD Eligibility

Dato- DXd does not meet the criteria for BTD.

BTD requires preliminary clinical evidence showing that the drug offers a substantial improvement over existing therapies on a clinically meaningful endpoint.

Although Dato- DXd achieved statistically significant improvements in PFS and ORR, the magnitude of benefit was modest – a median PFS extension of about 2 months, an absolute ORR increase of 13%, and no overall survival benefit (HR 1.01). This level of efficacy does not constitute a substantial improvement.

On the safety side, while Dato- DXd avoids the severe hematologic toxicities commonly associated with chemotherapy, it introduces new adverse events, and patients reported slightly worse overall side-effect bother (PGI- TT) compared to chemotherapy. As a result, the drug cannot be shown to offer a substantial improvement over available therapies.

3. AA vs. TA

3.1. AA Preferred in Oncology

When an oncology drug targets a serious or life- threatening disease with an unmet medical need, and a surrogate endpoint (ORR, DoR, pCR) or intermediate clinical

endpoint (PFS) can reasonably predict clinical benefit , and waiting for a final clinical endpoint (e.g., OS) would take too long or be unethical and the safety profile is acceptable, the FDA tends to prefer AA.

In oncology, typical scenarios that fit this pathway include: a single- arm trial showing an ORR significantly higher than historical controls, with durable DoR, and limited treatment options available – provided the surrogate endpoint has scientific predictive value.

3.2. TA Instead of AA

Dato- DXd received TA rather than AA.

Trial design: TROPION-Breast01 was a randomized, open-label, head-to-head Phase III trial directly comparing the drug to chemotherapy. It had dual primary endpoints of PFS and OS, and it met the PFS endpoint with a statistically significant improvement. This trial design itself aligns with traditional approval standards.

Endpoints rationale: The FDA concluded that although Dato- DXd showed no OS benefit, it also did not show a clear trend toward harm, and PFS endpoint with a statistically significant improvement. So PFS as a clinical endpoint supported traditional approval.

Pathway fit: The improvements in ORR and DoR for Dato- DXd were modest, and the sponsor had already completed a Phase III trial, so there was no need to rely on an early- data accelerated pathway.

Safety profile: While the drug avoids the hematologic toxicities of chemotherapy, it introduces new adverse events, and patients reported slightly worse side- effect bother compared to chemotherapy. The benefit- risk balance would struggle to meet the bar that AA sets for surrogate- endpoint benefit.

3.3. If the Trial Were Single- Arm with an ORR of ~36%, Would AA Be Appropriate?

Whether the FDA would approve depends on the DoR and the safety profile.

The AA pathway permits approval based on a surrogate endpoint like ORR from a single- arm trial, provided the ORR is significantly higher than historical controls and the DoR can reasonably predict clinical benefit. Under this hypothetical scenario, Dato- DXd's ORR (36%) is indeed higher than the chemotherapy arm (20–25%), and there is a clear unmet medical need for this disease. So a single- arm trial would meet the basic requirements for seeking AA.

However, when reviewing an AA application, the FDA places even greater emphasis on DoR durability and whether the safety profile is acceptable. If a single- arm trial also shows no meaningful improvement in DoR, it becomes difficult to predict subsequent clinical benefit. Moreover, Dato- DXd's safety profile is not superior to chemotherapy.

In summary, unless the DoR is very durable and the safety profile can be effectively managed, the FDA would likely refuse approval.

4. Benefit–Risk & Regulatory Decision-Making

4.1. Risks of AA

Withdrawal risk: If confirmatory trials fail to verify clinical benefit, the FDA may withdraw the approval.

Confirmatory trial failure or delay: Sponsors may be unable to complete confirmatory trials on time ,for example, due to patient enrollment difficulties or poor trial design. Even if completed, if the trial fails to meet its primary endpoint, the post- marketing requirement is not satisfied.

Regulatory uncertainty: The link between a surrogate endpoint and clinical benefit is not set in stone; a surrogate once considered reasonable may later be questioned by new data. This uncertainty creates challenges for both sponsors and regulators.

4.2. The Reason FDA Accept the Head- to- Head Trial Results for Dato- DXd

A key basis for the FDA's approval of Dato- DXd was that the TROPION- Breast01 trial directly compared Dato- DXd to active chemotherapy, not to a placebo.

The median PFS in the Dato- DXd arm was 6.9 months, meaning that patients gained about 2 months of additional progression-free survival on top of the benefit from standard treatment.

Second, the PFS results from TROPION- Breast01 were consistent across sensitivity analyses, investigator assessments, and various subgroups. The HR of 0.63 had a confidence interval entirely below 1.0, showing very high statistical significance – meaning the PFS benefit had to be robust. Moreover, the secondary endpoints (ORR and DoR) consistently supported the antitumor activity of Dato- DXd, further reinforcing the credibility of the PFS benefit.

Regarding OS, the FDA does not require a new drug to extend overall survival, especially in a head-to-head trial. In this trial, the OS hazard ratio was 1.01, no benefit, but also no clear trend toward harm. The FDA considered that, given a robust PFS benefit, a neutral OS result did not pose an obstacle to approval.

In summary, the FDA accepted traditional approval for Dato- DXd based on a robust PFS benefit from a head- to- head trial, no detrimental trend in OS, and consistency across secondary endpoints.

4.3. How Did Safety Influence the Decision

In the TROPION- Breast01 trial, the safety profiles of Dato- DXd and chemotherapy

were significantly different. The FDA concluded that each therapeutic option had different side effect profiles.

The main advantage of Dato-DXd is that it largely avoids the severe hematologic toxicities commonly seen with chemotherapy: the incidence of grade 3-4 neutropenia was only 1.6%, compared to 35% in the chemotherapy arm.

However, Dato-DXd also introduces adverse events that are uncommon with chemotherapy. Any-grade stomatitis occurred in 59% of patients; dry eye and keratitis occurred in 27% and 24%, respectively; interstitial lung disease (ILD) occurred in 4.2% of patients in the Dato-DXd arm, including one fatal case. In addition, patients reported slightly worse overall side-effect bother (PGI-TT) compared to chemotherapy.

Faced with these differences, the FDA applied an individualized benefit-risk assessment. On the one hand, the marked reduction in severe hematologic toxicity makes Dato-DXd a reasonable alternative for patients who cannot tolerate or wish to avoid myelosuppression. On the other hand, the new toxicities have all been addressed in the drug labeling through warnings and preventive measures (U.S. Food and Drug Administration, 2024a). The FDA explicitly stated that physicians can choose between Dato-DXd and chemotherapy based on each patient's individual circumstances.

Overall, the FDA therefore concluded that the benefit-risk balance for this drug is acceptable and approved it for marketing.

5. Strategic Regulatory Planning

5.1. Propose a regulatory strategy

Before launching the randomized Phase III TROPION-Breast01 trial, the sponsor had already obtained preliminary Phase I/II data. The disease is a serious,

life-threatening malignancy with a clear unmet medical need after endocrine therapy and chemotherapy. To gather more Phase I/II data, I found the multi-discipline review report and prescribing information on the FDA website.

Whether to apply for BT

According to multi-discipline review report (U.S. Food and Drug Administration, 2024b), in the TP01 study, patients receiving Dato-DXd at 6 mg/kg achieved a confirmed ORR of 26.8%, and the median DoR had not yet been reached. While these data are better than historical chemotherapy controls, the magnitude of improvement is not striking. At the same time, Phase I dose-finding showed that the 8 mg/kg dose (the MTD) had unacceptable toxicity, while the 6 mg/kg dose offered only limited efficacy improvement. Therefore, based on the Phase I/II data available at the time, the likelihood of obtaining BTd was low. Nevertheless, applying for BTd is low-cost and could facilitate early communication with the FDA, so it could be pursued as an exploratory application but not as a core strategic path.

AA vs. TA

AA would allow earlier marketing based on a surrogate endpoint, but it requires post-marketing confirmatory trials. In TP01, Dato-DXd did not show efficacy dramatically superior to historical chemotherapy, so the benefit-risk argument for AA might be weak. Moreover, if a subsequent confirmatory trial failed to verify clinical benefit, the accelerated approval would face withdrawal risk. In hindsight, the final Phase III trial did not show an OS benefit, so the AA pathway was not an ideal first choice.

In contrast, directly conducting a randomized, controlled Phase III trial to pursue TA was more solid. TA requires demonstration of clinical benefit based on adequate and well-controlled trials, with PFS as a primary endpoint. The final Phase III results showed a robust PFS benefit and no detrimental trend in OS.

Final strategy recommendation:

Prioritize the randomized Phase III trial (TA) while also applying for Fast Track designation and it offers rolling review and frequent FDA communication.

5.2. Alternative development strategy**5.2.1. Biomarker- enriched population**

According to the FDA Approval Summary (U.S. Food and Drug Administration, 2024a), although Trop-2 expression levels did not effectively predict efficacy in the TROPION-Breast01 trial, another ADC in the same class – fam-trastuzumab deruxtecan – has made breakthroughs in the HER2- low (IHC 1+ or IHC 2+/ISH-) and HER2- ultralow populations. About 60–70% of HR+/HER2- breast cancers are HER2- low or ultralow, and these patients still lack highly effective, low-toxicity treatment options after progression on endocrine therapy. Therefore, the alternative strategy would restrict the target population to patients with advanced HR+/HER2- low or HER2- ultralow breast cancer who have progressed after at least one prior line of endocrine therapy and one prior line of chemotherapy in the metastatic setting.

5.2.2. Smaller- scale single- arm Phase II trial

A single-arm, open-label design would be used, eliminating the need for a concurrent randomized control group, thereby substantially reducing sample size and follow-up time. Under an optimal Simon two-stage design, total enrollment would be approximately 95 patients. The primary endpoint would be ORR assessed by blinded independent central review (BICR); secondary endpoints would include DoR, PFS, safety, and patient-reported outcomes. This size is far smaller than the 732 patients in TROPION-Breast01, significantly shortening the enrollment period and lowering costs.

5.2.3. Using a surrogate endpoint to apply for AA

After completing the single-arm Phase II trial, if the confirmed ORR is $\geq 35\%$, the

median DoR is ≥ 6 months, and the safety profile is acceptable , then can apply AA. PR and FT could also be stacked on top, allowing rolling submissions to compress the review timeline.

5.2.4. Post- marketing confirmatory trial

After approval, a confirmatory trial would be conducted in the same patient population, with PFS as the primary endpoint and OS as a key secondary endpoint. Before finishing Phase II, a meeting with the FDA would be held to align on the population, endpoints, and confirmatory trial design, thereby reducing regulatory risk. If the confirmatory trial fails, a withdrawal procedure would be initiated.

5.2.5. Advantages and risks

Advantages: Time to market could be shortened; R&D costs could be reduced; and the strategy fully leverages FDA's AA and FT mechanisms.

Risks: Single- arm trials lack a randomized control and may overestimate efficacy; if the confirmatory trial fails after AA, the approval may be withdrawn; and changes in regulatory policy could introduce uncertainty.

6. Bonus

6.1 Evaluate dose optimization addressed

Dato- DXd's dose selection strategy was already quite sound by conventional standards. However, when measured against the stricter criteria of Project Optimus, there is still some room for improvement.

Dose exploration and selection:

According to multi- discipline review report (U.S. Food and Drug Administration, 2024b).

In the Phase I TROPION- PanTumor01 trial, Dato- DXd was studied over a broad dose

range of 0.27–10 mg/kg (every 3 weeks) in both dose-escalation and dose-expansion phases. During the dose-expansion part, patients in the NSCLC cohort received Dato- DXd at 4 mg/kg, 6 mg/kg, or 8 mg/kg

The sponsor did not blindly choose the MTD (8 mg/kg). They performed a systematic comparison: 8 mg/kg gave the strongest efficacy but higher toxicity and was intolerable; 4 mg/kg was best tolerated but had insufficient efficacy; 6 mg/kg was clearly identified by the FDA as the highest tolerable dose that provided an acceptable benefit-risk profile. Moreover, exposure-response analysis showed a positive correlation between Dato- DXd exposure and ocular surface toxicity (OST). As a result, the FDA recommended a dose cap of 540 mg for patients weighing ≥ 90 kg (U.S. Food and Drug Administration, 2024c).

Methodological support:

The sponsor used a model-informed drug development (MIDD) approach, integrating population PK and exposure-response analyses to demonstrate that exposure was positively correlated with several safety endpoints, supporting the choice of 6 mg/kg (U.S. Food and Drug Administration, 2024b).

Major gaps by Project Optimus standards:

Under Project Optimus criteria, it still has the following gaps: dose selection was based primarily on non-randomized exploration in Phase I and model simulations, without simultaneously including a low-dose randomized arm for head-to-head comparison. In addition, dose optimization was largely locked in early stages; once the drug entered Phase III, only a single dose (6 mg/kg) was tested, and no further optimization was pursued in later phases.

6.2. Improvement strategy

Conduct a randomized dose-comparison trial in Phase II: After identifying candidate doses from TROPION- PanTumor01, design a randomized Phase II trial that assigns

patients to three arms – 4 mg/kg, 6 mg/kg, and 8 mg/kg – to directly compare ORR, DoR, and safety profiles across doses.

Include multiple doses or use an adaptive design in the registration trial: If the randomized Phase II comparison shows that 5 mg/kg or 4 mg/kg is non-inferior in efficacy and has better tolerability, that dose could be added as a third arm in the Phase III trial.

References:

U.S. Food and Drug Administration. (2024a). *Approval summary: Datopotamab deruxtecan (Dato-DXd) for HR+/HER2- metastatic breast cancer.*

U.S. Food and Drug Administration. (2024b). *Multidiscipline review report: Datopotamab deruxtecan.*

U.S. Food and Drug Administration. (2024c). *Datopotamab deruxtecan prescribing information.*